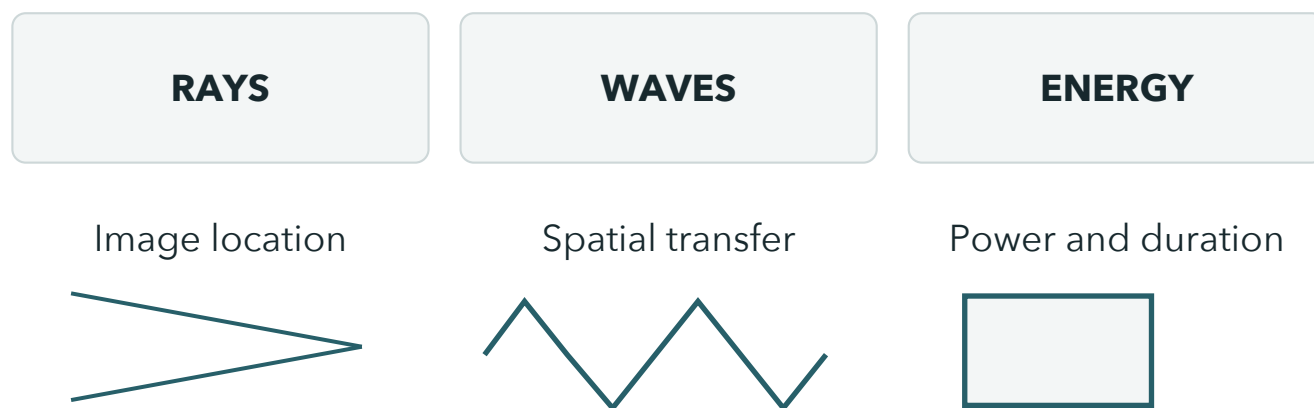


Optics, refraction and instrument physics

Part 1 | Optical reasoning needs the correct model

Ray, wave and energy descriptions answer related but different questions.



Geometric rays, a spatial wave pattern and a power-time pulse represent three complementary models.

Model

Feature	Predicts	Typical limitation
Paraxial rays	Image location and magnification	Not diffraction
Wave optics	Interference and spatial transfer	Needs coherence/aperture assumptions
Radiometry	Physical power and energy	Not perceived brightness alone

Separate surface power from propagation

Distance between elements can change the vergence arriving at the next element.

Original +8 D then +4 D, 50 mm apart

Step	Vergence
Collimated input to first lens	+8 D after lens
Travel through air	$8/(1-0.05 \times 8) = +13.33$ D
Second lens	$13.33 + 4 = +17.33$ D

Reference-plane result

Effective focal length is referenced to a principal plane; a vertex focal distance is referenced to a lens surface. If the principal plane is displaced, adding the focal distance to its signed coordinate locates the image. Omitting that displacement changes the answer even for an infinitely distant object.

Cylinder and image-size effects are different

A prescription can alter focal quality and magnification through separate components.

Optical effect

Feature	Main comparison	Consequence
Regular astigmatism	Two principal powers	Two orthogonal focal lines
Sphere adjustment	Equal power added to both	Interval in vergence unchanged
Spectacle shape factor	Thickness/index/front curve	Size can differ at equal prescription

Worked size reasoning

With equal power factors, a shape factor of 1.03 versus 1.01 gives a relative size ratio of $1.03/1.01$, about 1.020. Correct focus and equal perceived image size are therefore separate goals. Retinal deformation can further limit what a uniform optical size correction achieves.

Image quality involves contrast as well as cutoff

An aperture can reduce aberrations but impose greater diffraction at very small sizes.

Distinction

Feature	Quantity	Meaning
Resolution limit	Highest useful spatial detail	Not all contrast performance
MTF	Transferred modulation by frequency	Curve rather than single endpoint
Photopic flux	$683 \times \text{watts} \times V(\lambda)$	Visual weighting differs by spectrum

Worked photometry

A 2 mW monochromatic source with $V=0.25$ gives $683 \times 0.002 \times 0.25 = 0.3415$ lumen. Equal radiant watts at $V=1$ would give 1.366 lumens. Neither luminous flux alone nor a diffraction cutoff establishes the patient's complete visual performance.

Instrumentation selects information

Sectioning, field and dose should be described independently.

Technique

Feature	Selective principle	Interpretive consequence
Confocal	Spatial rejection of out-of-focus light	Optical sectioning
OCT	Coherence-based depth discrimination	Axial structure
Indirect ophthalmoscopy	Condensing-lens image	Field-magnification trade-off
Pulsed laser	Energy over area and time	Duration changes tissue interaction

Worked dose

A uniform 0.05 J pulse over 0.02 cm² gives 2.5 J/cm² radiant exposure. Its irradiance still depends on pulse duration. This separates energy density from power density before any biological conclusion about the target is made.